

EXPERIMENT-2

ROLL NO:240701221

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USER INTERFACE AND DESIGN

CHUNKING AND RECALLING VISUAL ELEMENTS



FRAME 1:

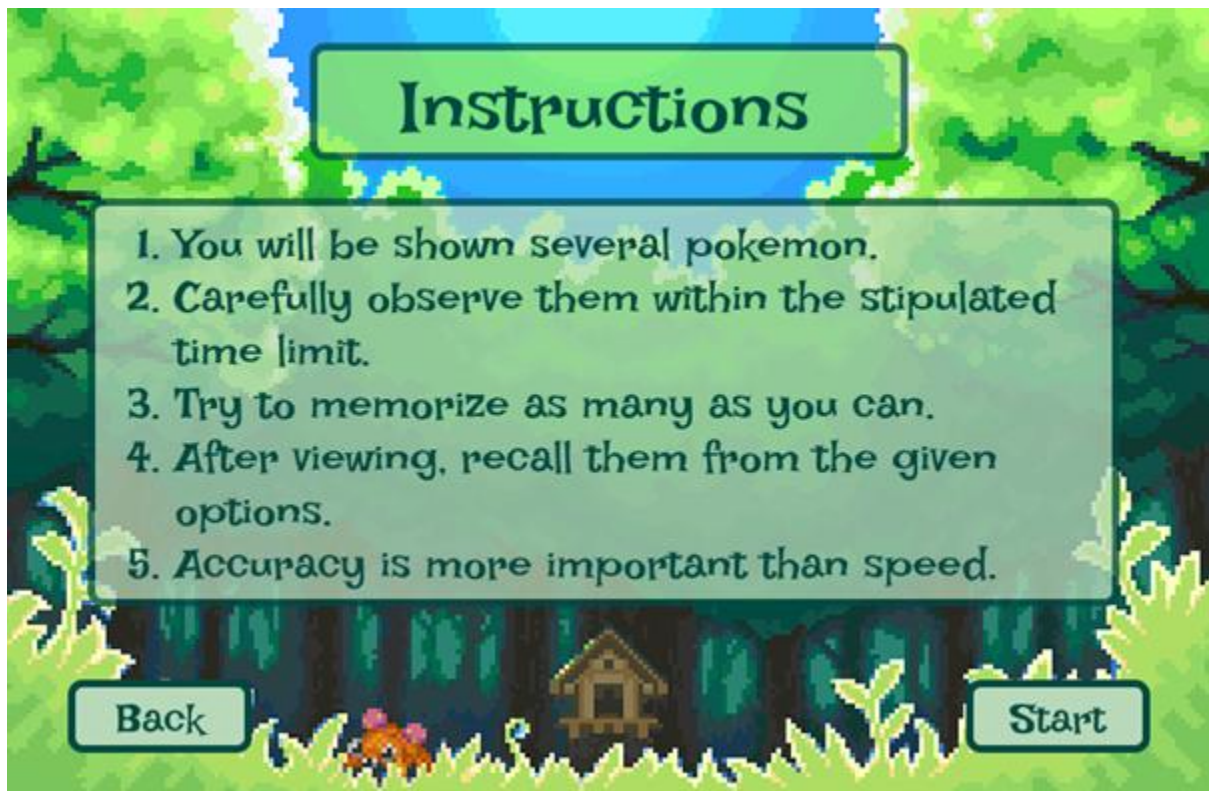
TITLE / START SCREEN

Chunking Analysis of the Title Screen

Chunking at the entry level helps users immediately **understand the game structure** and available actions without overwhelming them.

- 1. Purpose of the Screen**
 - Acts as the entry point to the game.
 - Introduces the game theme and sets user expectations.
 - Allows users to choose between starting the game or viewing instructions.
- 2. Clear Grouping of Actions**
 - The game title is visually separated from action buttons.
 - Two primary options are provided:
 - o “Start” → Begin gameplay.

- o “How To” → View instructions before playing.
 - This separation prevents decision overload for first-time users.
3. **Visual Hierarchy and Design Elements**
 - Large wooden signboard displaying “Pokémon Battleground” immediately draws attention.
 - Buttons are centrally aligned and clearly labeled.
 - Forest-themed pixel background establishes consistency across all future screens.
 4. **Cognitive & UX Benefits**
 - Minimal elements reduce cognitive load.
 - Clear action paths improve usability and onboarding.
 - Familiar Pokémon-style visuals increase engagement and comfort.



FRAME 2:
INSTRUCTION PAGE

Chunking Analysis of the Instruction Page

Chunking is a **cognitive strategy** that breaks down information into smaller, manageable units, making it easier to process and retain. The Pokémon Battleground instruction page applies chunking effectively in the following ways:

1. **Clear and Sequential Numbering**
 - The instructions are presented in numbered steps (1–5), ensuring a logical flow.
 - Each step focuses on a single action, reducing confusion and cognitive overload.
2. **Logical Grouping of Information**
 - Exposure Phase (Steps 1–2) → Explains that multiple Pokémon will be shown within a time limit.

- Memorization Strategy (Step 3) → Encourages users to focus on remembering as many Pokémon as possible.
- Recall Phase (Step 4) → Informs users that they must recall Pokémon from given options.
- Performance Emphasis (Step 5) → Highlights that accuracy is prioritized over speed.

3. Visual Hierarchy and Design Elements

- Prominent “Instructions” header → Clearly establishes the purpose of the page.
- Semi-transparent instruction panel → Separates text from the forest background for readability.
- Consistent Pokémon-themed visuals → Maintains immersion without distracting from the text.
- Clearly placed “Back” and “Start” buttons → Provide intuitive navigation options.

4. Time Constraint Reinforcement

- The mention of a stipulated time limit sets expectations early.
- Prepares users mentally for the fast-paced memory challenge ahead.

5. Simplicity and Clarity

- Instructions are short, direct, and written in simple language.
- The tone is instructional yet friendly, suitable for a game-based learning environment.



FRAME 3:
CHUNKING PHASE

Analysis of the Pokémon Battleground – Memorization Screen

This screen represents the **Chunking (Encoding) Phase**, where users visually process and memorize Pokémon within a limited time.

1. Purpose of the Screen

- Serves as the memory encoding stage where users observe multiple Pokémon.
- Encourages mental grouping (chunking) to improve short-term memory retention.

2. Key Elements and UI Components

Countdown Timer (Left Side – “00:10”)

- Displays the remaining time clearly, creating urgency.
- Encourages rapid visual scanning and focused attention.

Progress Indicator (Top Center)

- A segmented progress bar visually represents remaining time.
- Reinforces the time constraint without relying only on numbers.

Pokémon Grid

- Pokémon are displayed in a structured grid layout.
- Grid arrangement supports chunking strategies such as:
 - o Grouping by Pokémon type (fire, water, grass).
 - o Categorizing by color schemes or visual similarity.
 - o Remembering evolutionary patterns or familiar starters.

Instruction Tooltip (Right Side)

- Reminds users of the objective: “Try to remember as many Pokémon as you can within the time limit.”
- Reinforces task goals without interrupting gameplay.

3. How the Chunking Phase Works

4. Users scan the Pokémon grid and identify familiar or related characters.

5. Time pressure encourages chunk-based memory strategies rather than individual memorization.

6. Once time expires, the game transitions automatically to the recall phase.

7. Cognitive and UX Benefits

- Improves memory retention through visual grouping.
- Reduces cognitive load by presenting items in a predictable grid structure.
- Familiar Pokémon designs enhance recall through prior knowledge.



FRAME 4:
RECALL PHASE

Analysis of the Pokémon Battleground – Selection Screen

This screen represents the **Recall Phase**, where users retrieve stored information and identify Pokémon seen previously.

1. Purpose of the Screen

- Tests short-term memory by requiring users to recognize previously shown Pokémon.
- Measures the effectiveness of chunking during the memorization phase.

2. Key Elements and UI Components

Instruction Prompt

- “Select the Pokémon that you remember from the previous slide!”
- Clear, action-oriented instruction reduces ambiguity.

☐ Pokémon Selection Options

- A smaller set of Pokémon is shown, mixing correct answers with distractors.
- Users must differentiate between seen and unseen Pokémon.
- Visual selection boxes clearly indicate selectable items.

☒ Action Buttons

- “Retry” button → Allows users to restart the recall process.
- “Submit” button → Finalizes selections and proceeds to scoring.

3. How the Recall Phase Works

4. Users mentally retrieve Pokémon from memory.

5. They compare recalled images with the presented options.
6. Distractor Pokémon test accuracy and resistance to false memory.
7. Submission confirms the recall attempt.
8. Cognitive and UX Benefits
 - Strengthens retrieval practice, improving memory consolidation.
 - Use of distractors evaluates true recall rather than guessing.
 - Simple interaction design keeps focus on cognition, not controls.



FRAME 5:
RESULT PAGE

Analysis of the Pokémon Battleground – Score & Feedback Screen

This screen represents the Feedback Phase, where users receive performance results and next-step options.

1. Purpose of the Screen
 - Communicates recall accuracy clearly and immediately.
 - Encourages continued engagement through rewards and replay options.
2. Key Elements & UI Components

Final Score Display

- “Final Score” header clearly marks completion of the task.
- Numerical score (e.g., 24/24) is prominently displayed.
- Text feedback (“That’s a full score, congrats are in order!”) reinforces positive achievement.

★ Visual Rewards

- Star icons and Poké Ball graphics serve as motivational reinforcement.
- Visual celebration increases satisfaction and replay value.

■ Action Buttons

- “Restart” → Replay the game.
- “Continue” → Move to the next challenge or level.
- “Exit” → Leave the game.

3. How This Phase Works

4. The system evaluates recall accuracy.

5. The final score is displayed with encouraging feedback.

6. Users choose their next action based on motivation and performance.

7. Cognitive & UX Benefits

- Immediate feedback supports learning and self-assessment.
- Positive reinforcement boosts confidence and motivation.
- Multiple exit paths respect user autonomy and reduce frustration.

PROTOTYPE LINK: [Chunking – Figma](#)